

Construction Company ERP

Full Product, Workflow, Calculation, Portal, API & Agentic Implementation Specification

Version: 1.0

Target: Production-grade construction company operating system

Architecture benchmark: 2026 construction-management and agentic-software standards

1. PRODUCT OBJECTIVE

The objective is **not simply to build a project-management application**.

The system should become the complete operating system for a construction company covering:

- Company management
- Employees and organizational structure
- CRM and clients
- Consultants
- Main contractor operations
- Subcontractors
- Suppliers
- Tendering and estimation
- Projects
- Project teams
- Site organization
- Buildings/floors/zones
- BOQ and quantity takeoff
- Materials and inventory
- Procurement
- Site execution
- Daily works
- Inspections
- Consultant observations
- RFIs
- Submittals

- Drawings and documents
- QA/QC
- HSE
- Scheduling
- Cost control
- Client contracts
- Subcontracts
- Variations
- Invoicing
- Collections
- Supplier/subcontractor payments
- Equipment
- Labour
- Handover
- Warranty
- Company reporting
- Project reporting
- Notifications/reminders
- AI assistance
- REST/API integration
- MCP connectivity
- Fully auditable agent actions

The current system already contains a meaningful execution foundation including project setup, phases, teams, milestones, BOQ, work orders, site reporting, QHSE, RFIs, submittals, documents and property sales.

The target is to **expand the existing system rather than create disconnected modules around it.**

2. CORE ARCHITECTURAL PRINCIPLE

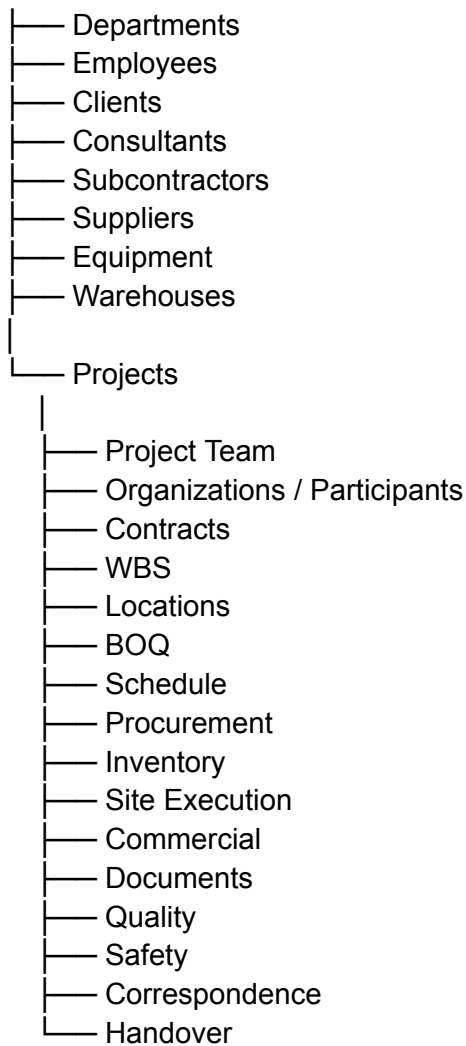
Everything should answer:

Which company? → Which project? → Which location? → Which work package? → Which responsible party? → What quantity/cost? → What approval state? → What action is required?

The basic data structure should therefore be:

Company

|



Do not duplicate information separately for consultant/client/subcontractor apps.

They should all use the **same project database** with different permissions and interfaces.

3. MULTI-ORGANIZATION MODEL

Create a first-class **Organization** object.

Organization types:

Internal Company

Client / Owner

PMC / Construction Manager
Consultant
Architect
Structural Consultant
MEP Consultant
QS Consultant
Subcontractor
Supplier
Labor Contractor
Testing Laboratory
Authority
Broker
Other Third Party

Each organization contains:

- Legal name
- Trade name
- Registration
- Tax number
- Address
- Contacts
- Bank information
- Documents
- Certifications
- Insurance
- Commercial registration
- VAT registration
- Trade/speciality
- Qualification status
- Performance history
- Blacklist status
- Active/inactive
- Associated projects
- Users

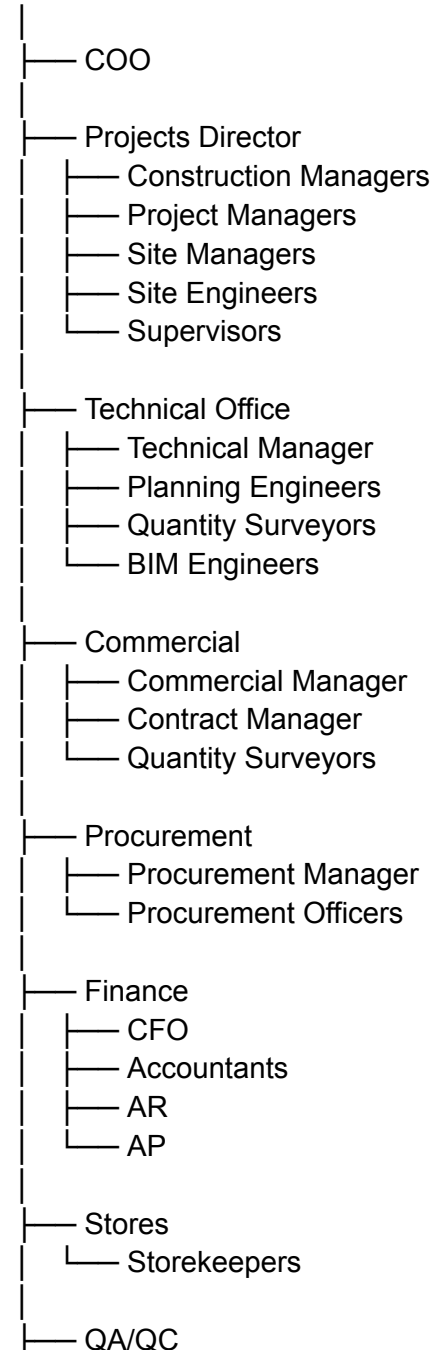
Individual users belong to organizations.

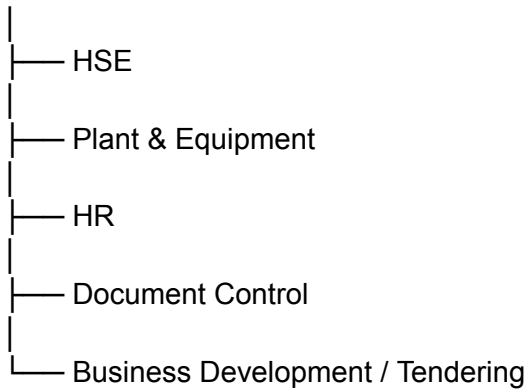
4. COMPANY ORGANIZATIONAL STRUCTURE

The ERP should support the full company hierarchy.

Example:

CEO / Managing Director





The hierarchy must be configurable.

5. ROLE AND PERMISSION ENGINE

Do not hard-code only:

Admin
Manager
Employee

Permission should be calculated using:

Company
+
Organization
+
Project
+
Role
+
Module
+
Record
+
Action

Actions:

View
Create
Edit
Delete
Submit
Review
Approve
Reject
Return
Assign
Close
Reopen
Export
Download
See Financial Values
See Internal Costs
See Client Price
See Subcontractor Price

For example:

Project Manager

Project A

RFIs	FULL
Submittals	FULL
Budget	VIEW
Internal Cost	VIEW
Payments	VIEW

Consultant

Project A

RFIs	REVIEW + RESPOND
Submittals	REVIEW + APPROVE
Budget	NONE
Internal Cost	NONE

Client

Project A

Progress	VIEW
Milestones	VIEW
Photos	VIEW
Variations	REVIEW/APPROVE
Internal Cost	NONE

6. PROJECT CREATION MUST BECOME A WIZARD

Creating a project should configure the entire project environment.

Step 1 — Basic information

Fields:

- Project number
- Arabic name
- English name
- Project type
- Client
- Contract type
- Location
- GPS location
- Country
- Currency
- Tax configuration
- Contract value
- Project budget
- Planned start
- Planned finish
- Defects liability period
- Retention rules
- Advance payment
- Status

Step 2 — Project physical structure

Configure:

Project

→ Sites

→ Buildings / Blocks

→ Zones

- Floors
- Areas / Apartments / Rooms

For infrastructure:

- Project
- Sections
 - Zones
 - Chainage
-

Step 3 — Project WBS

Example:

- 1 Preliminaries
- 2 Substructure
- 3 Superstructure
- 4 Architectural
- 5 MEP
- 6 External Works
- 7 Testing & Commissioning
- 8 Handover

Allow import from template.

Step 4 — Project team

While creating the project, assign:

- Project Director
- Construction Manager
- Project Manager
- Site Manager
- Planning Engineer
- QS
- Technical Office
- QA/QC
- HSE

- Procurement responsible
- Accountant
- Storekeeper
- Document Controller
- Engineers
- Supervisors

Each assignment automatically affects permissions, notifications and dashboards.

The existing project module already supports assigning project-team members and roles, but this should now become part of structured project creation and access control.

Step 5 — External project participants

Add:

- Client
- PMC
- Lead consultant
- Architect
- Structural consultant
- MEP consultant
- QS consultant
- Primary subcontractors
- Suppliers

Invite users immediately or later.

Step 6 — Approval matrix

Example:

RFI:

Site Engineer

→ PM

→ Consultant

Submittal:

Technical Office

→ PM

→ Consultant

Purchase > 100k:

Procurement

→ PM

→ Commercial

→ GM

Variation > 500k:

PM

→ Commercial Manager

→ Director

→ Client

Step 7 — Cost structure

Import/create:

- Cost codes
- BOQ
- Budget
- CBS
- Work packages

Step 8 — Schedule template

Import:

- Project schedule
- WBS
- activities
- milestones

or create manually.

Step 9 — Project document structure

Automatically create standard folders/registers.

Step 10 — Notifications

Configure:

- SLA
 - reminders
 - escalation rules
 - email/push/in-app rules
-

Step 11 — Finish

System generates the complete **Project Workspace** automatically.

There must never be a half-created project.

7. PROJECT DIGITAL FILE

Every project needs one unified digital project file.

Navigation:

- 01 Overview
- 02 Team
- 03 Participants
- 04 Locations
- 05 WBS & Schedule
- 06 BOQ / Quantities
- 07 Budget & Cost
- 08 Contracts
- 09 Procurement
- 10 Materials & Inventory
- 11 Subcontractors
- 12 Work Packages
- 13 Site Execution
- 14 Daily Reports

- 15 Consultant Visits
- 16 Observations
- 17 RFIs
- 18 Submittals
- 19 Inspections
- 20 QA/QC
- 21 HSE
- 22 Drawings
- 23 Documents
- 24 Correspondence
- 25 Meetings
- 26 Variations
- 27 Payment Certificates
- 28 Client Invoices
- 29 Supplier Invoices
- 30 Photos
- 31 Equipment
- 32 Labour
- 33 Reports
- 34 Handover
- 35 Warranty
- 36 Audit Log

One project = one complete file.

8. LOCATION-BASED PROJECT CONTROL

Every important project object must optionally contain:

- Site
- Building
- Zone
- Floor
- Area
- Room
- Work Package
- WBS Activity

Example:

Project: Tower Development
Site: Main Site
Building: Tower A
Floor: Floor 8
Zone: East
Room: Apartment 803
Activity: Ceramic Flooring

Then the system can answer exactly what happened there.

9. QUANTITY MODULE

This is a critical enhancement.

Every BOQ item receives a physical quantity allocation.

Example:

Location	Planned	Executed	Approved	Remaining
Basement	1,250 m ³	1,250	1,250	0
Ground	620	620	600	0
Floor 1	510	400	380	110
Floor 2	510	100	80	410

Calculations:

Remaining Quantity
=
BOQ Quantity - Executed Quantity

Certified Quantity
=
Consultant Approved Quantity

Physical Progress %

=
Approved Quantity / Planned Quantity × 100

Project totals roll upward:

Area
→ Floor
→ Building
→ Site
→ Project

The current work-order system already supports reporting and approving quantities against BOQ items.

The new design must add the location and cost dimensions.

10. MATERIAL RECIPE / CONSUMPTION ENGINE

Activities should support standard material recipes.

Example:

1 m³ Reinforced Concrete:

Concrete C35	1.00 m ³
Rebar	125 kg
Binding Wire	1.5 kg
Formwork	3.8 m ²
Spacer Blocks	10 pcs

Include configurable:

- wastage %
- conversion factors
- alternative materials
- project-specific ratios

If Floor 4 requires 500 m³:

Concrete:

$$500 \times 1.00 = 500 \text{ m}^3$$

Rebar:

$$500 \times 125 = 62,500 \text{ kg}$$

Binding Wire:

$$500 \times 1.5 = 750 \text{ kg}$$

System calculates requirements automatically.

11. FLOOR / LOCATION DASHBOARD

Every floor/location should have its own dashboard.

Display:

Progress

- planned %
- actual %
- approved %
- schedule status

Quantities

- planned
- executed
- certified
- remaining

Materials

- required
- reserved
- issued
- consumed
- returned

- waste
- remaining

Site

- active work
- manpower
- equipment

Consultant

- inspections
- comments
- open observations

Quality

- WIR
- MIR
- NCR
- tests

Information

- drawings
- RFIs
- submittals

Photos

- latest progress photos

Commercial

- budget
- actual cost
- committed cost
- forecast

12. MATERIAL PLANNING

Calculate:

Required Quantity

- Consumed

= Remaining Requirement

Then:

Remaining Requirement

- Stock Available

- Confirmed Incoming

=

Procurement Requirement

Material planning should also consider schedule dates.

Example:

Material: Blockwork

Need next 30 days 45,000

Stock 12,000

Confirmed PO 15,000

Shortage 18,000

Required date 25 Sep

Click shortage:

Tower A / Floor 5 6,500

Tower A / Floor 6 5,500

Tower B / Floor 2 6,000

13. INVENTORY MODULE

Warehouses:

Central Warehouse

Project Warehouse

Site Store

Temporary Store
Floor/Zone Holding Area

Every transaction should create a stock movement.

Movement types:

- GRN
- Issue
- Return
- Transfer
- Adjustment
- Waste
- Damage
- Supplier return

Calculation:

Physical Stock

=

Opening

+ Receipts

+ Returns

+ Transfers In

- Issues

- Transfers Out

- Waste

Also track:

Available Stock

=

Physical Stock - Reserved Stock

The current guide allows warehouse material use through work orders, but site delivery information is still largely free text.

That should become fully transactional.

14. AUTOMATIC REORDERING

Material master fields:

- Minimum stock
- Maximum stock
- Safety stock
- Reorder point
- Lead time
- Daily usage
- Preferred supplier
- Purchase unit
- order multiple
- latest rate
- framework price

Calculate:

Reorder Point

=

Lead-Time Consumption + Safety Stock

Forecast:

Projected Inventory

=

Current Stock

+ Confirmed Incoming

- Reserved Quantity

- Planned Consumption

When:

Projected Inventory $\leq$ Reorder Point

trigger replenishment.

Levels:

Mode A

Automatically create Purchase Requisition.

Mode B

Automatically create draft PO for approved framework supplier.

Mode C

Automatically issue PO below configured financial authority.

Example:

Auto PO $\leq$ EGP 20,000

20,001–100,000

→ Procurement Manager

100,001–500,000

→ Procurement + PM

>500,000

→ Commercial Director

Never allow uncontrolled automatic purchasing.

15. PROCUREMENT MODULE

Workflow:

Material Requirement

↓

Purchase Requisition

↓

Approval

↓

RFQ

↓

Supplier Quotations

↓

Technical Comparison

↓

Commercial Comparison

↓

Selection

↓

Purchase Order



Delivery



MIR



GRN



Inventory



Supplier Invoice



Payment

Documents:

- PR
- RFQ
- quotation
- comparison sheet
- technical evaluation
- commercial evaluation
- approval form
- PO
- delivery note
- MIR
- GRN
- supplier invoice
- payment voucher
- return note

Calculations:

Total Purchase

=

Qty × Unit Rate

plus:

- discount
- VAT
- freight
- customs where applicable

- retention
 - advance
 - other charges
-

16. SUPPLIER PORTAL

Supplier sees:

- RFQs
- quotation submission
- POs
- delivery dates
- documents
- required certifications
- invoices
- payment status

No internal company financial information.

17. SUBCONTRACTOR MODULE

Subcontractors become first-class companies.

Master data:

- classification
- trade
- qualification
- documents
- insurance
- certifications
- bank details
- performance
- blacklist
- projects

Each subcontract contains:

Package
BOQ
Original Amount
Approved Variations
Revised Amount
Executed
Certified
Retention
Advance
Advance Recovery
Paid
Outstanding

Calculation:

Revised Subcontract

=

Original Amount + Approved Changes

18. SUBCONTRACTOR PORTAL

Subcontractor sees:

- scope
- BOQ
- drawings
- work packages
- programme
- RFIs
- submittals
- WIR
- MIR
- NCR
- consultant observations
- progress quantities
- manpower
- equipment
- variation requests
- payment applications
- payment status

- snags
 - warranty tasks
-

19. CLIENT CONTRACT

Maintain:

- contract amount
- contract type
- BOQ/SOV
- commencement date
- completion date
- retention
- advance
- performance bond
- advance guarantee
- insurance
- liquidated damages
- taxes
- variations
- extensions
- payment terms

Documents:

- signed contract
 - LOA
 - BOQ
 - drawings
 - conditions
 - addenda
 - guarantees
 - insurance
 - amendments
-

20. COST CONTROL

The existing project display currently uses collected payments and expenses to derive project profit.

For construction control, that must become a much richer commercial model.

Revenue

Original Contract
+ Approved Client Variations
=
Revised Contract Value

Then:

- Work Executed
- Certified
- Invoiced
- Collected
- Retention Receivable
- Outstanding AR

Cost

Original Budget
+ Budget Adjustments
=
Current Budget

Then:

Committed Cost
Actual Cost
Accrued Cost
Forecast Cost to Complete
Forecast Final Cost

Margin

Forecast Profit
=
Forecast Revenue - Forecast Final Cost

Forecast Margin %

=

Forecast Profit / Forecast Revenue × 100

Oracle's current Primavera Unifier platform similarly treats cost management, controls, documentation and project processes as connected enterprise functions rather than merely comparing cash receipts to recorded expenses.

21. CLIENT INVOICING / ACCOUNTS RECEIVABLE

Invoice creation:

- client
- project
- contract
- valuation
- billing period
- certified amount
- retention
- advance recovery
- taxes
- deductions
- previous certified
- current certified

Calculation example:

Gross Certified

- Retention

- Advance Recovery

- Other Deductions

+ Tax

=

Invoice Amount

Statuses:

Draft

Approved
Issued
Partially Paid
Paid
Overdue
Cancelled

Reminder engine:

7 days before due date
Due today
7 days overdue
14 days overdue
30 days overdue

Escalate according to configured rules.

22. ACCOUNTS PAYABLE

Supplier/subcontractor invoice flow:

Invoice Received
↓
PO/Subcontract Match
↓
Quantity Verification
↓
Commercial Verification
↓
Approval
↓
Due
↓
Payment

Support three-way matching:

PO
↔

GRN

↔

Supplier Invoice

Exceptions are flagged automatically.

23. FINANCE MODULE

Company-wide finance dashboard:

- AR
- AP
- cash
- project profitability
- invoices
- collections
- supplier liabilities
- subcontract liabilities
- retention
- VAT/tax
- cash flow

Optionally integrate later with accounting packages rather than recreate a full statutory general ledger if not required.

Every transaction should carry:

Company
Project
Cost Code
Department
Supplier/Client
Date
Currency
Tax
Approver
Attachments

24. CONSULTANT PORTAL

Consultants get real login credentials.

Their interface should be significantly simpler than the internal ERP.

Home:

- Project Summary
- Milestones
- Overall Progress
- Today's Inspections
- Pending Reviews
- New Work Ready for Inspection
- Open Observations
- RFIs
- Submittals
- Latest Drawings
- Site Visits
- Daily Progress
- Messages
- Notes

Autodesk currently uses enhanced RFI workflows with coordinators and expert reviewers rather than only basic sender/receiver flows, which is a useful model for this portal.

25. CONSULTANT DAILY DASHBOARD

Display:

TODAY

- 4 WIRs awaiting inspection
- 2 MIRs
- 3 RFIs requiring reply
- 5 Submittals awaiting review
- 2 previous observations awaiting closure
- 1 planned site visit

The consultant does not need to navigate the full ERP.

26. CONSULTANT SITE VISIT

Fields:

- visit number
- visitor
- organization
- discipline
- project
- site
- building
- floor
- exact location
- date/time
- attendees
- purpose
- activities inspected
- drawings referenced
- photos
- notes

The existing site-visit module already supports visitor information, notes, follow-up items and photographs.

Expand it into a formal observation process.

27. CONSULTANT OBSERVATION

Each comment must become a record.

Example:

OBS-000341

Project:

Tower Development

Tower:

A

Floor:

06

Area:

Apartment 603

Raised By:

Structural Consultant

Priority:

High

Comment:

Rebar spacing differs from approved drawing.

Reference:

STR-601 Rev 04

Attach photos.

28. OBSERVATION WORKFLOW

Consultant Creates Observation



Project Manager Notification



PM Acknowledges



PM Assigns Responsible Engineer/Subcontractor



Rectification



Before/After Photos Uploaded



PM Reviews



Submitted to Consultant

↓

Consultant Approves or Rejects

↓

Closed

Rejection restarts correction.

Every status change retains:

- user
 - company
 - timestamp
 - comments
 - photographs
 - attachments
-

29. PHOTO MANAGEMENT

Every photo should contain metadata:

Project

Building

Floor

Location

Work Package

Uploader

Organization

Date

Linked Record

Enable annotations:

- arrow
- circle
- box
- text
- measurements

Maintain:

BEFORE
DURING
AFTER

30. STICKY NOTES / PERSONAL NOTES

Every dashboard should support lightweight notes.

Types:

Personal note

Only user sees it.

Project note

Shared with selected project team.

Location note

Attached to floor/location.

Record note

Attached to RFI/submittal/etc.

Functions:

- sticky/pin
- colour/category
- due date
- reminder
- convert to action
- assign person

Sticky notes must **not replace formal project correspondence**.

31. CLIENT PORTAL

Client login should be extremely clean.

Dashboard:

- project progress
- planned vs actual
- milestones
- photographs
- schedule summary
- approved changes
- pending client decisions
- certified values
- invoices
- payments
- key documents
- major RFIs if permitted
- reports
- handover status

They should never accidentally see:

- internal margins
- subcontractor rates
- supplier rates
- confidential correspondence

32. CLIENT PORTAL MUST ALWAYS WORK

Critical product requirement:

The Client Dashboard route must render even if:

- project has no client assigned
- client has no portal user
- no invoices exist
- no progress exists

Use proper empty states:

No client assigned yet

[Assign Client]

Never:

null
undefined
500
blank page

Same for:

- consultant portal
- subcontractor portal
- supplier portal
- every dashboard

Components must accept empty datasets.

33. DEMO / PREVIEW MODE

Admin should be able to:

Preview as Client
Preview as Consultant
Preview as Subcontractor
Preview as Site Engineer

without creating temporary fake accounts.

This allows QA and demonstrations.

34. WORK PACKAGE ENGINE

Create a reusable Work Package entity.

Example:

WP-ARCH-004
Tower A – Floor 04 Finishes

It connects:

- WBS
- BOQ
- location
- subcontractor
- schedule
- materials
- drawings
- RFIs
- submittals
- inspections
- budget
- quantities

Dashboard:

Progress	68%
Schedule variance	-5 days
Budget consumed	63%
Material readiness	92%
Open inspections	4
Open consultant comments	2
NCR	1

35. SCHEDULING

Add:

- WBS
- activities
- predecessors
- successors
- FS / SS / FF / SF
- lags
- calendars
- baseline

- planned start
- planned finish
- actual start
- actual finish
- remaining duration
- progress %
- critical path
- float
- resources

Views:

- Gantt
- 2-week lookahead
- 4-week lookahead
- 6-week lookahead
- milestones
- delayed activities

Support eventual import/export with Primavera/MS Project rather than trying to recreate all Primavera capabilities initially.

36. SCHEDULE CALCULATIONS

Schedule Variance

=

Actual Progress - Planned Progress

Earned-value options:

$SPI = EV / PV$

$CPI = EV / AC$

Show only where project controls use earned value.

37. DAILY SITE OPERATIONS

Site engineer home screen:

TODAY

Activities
Inspections
Materials
Deliveries
Manpower
Equipment
Consultant comments
Safety items
Drawings
Daily Report
Photos

Mobile-first.

38. DAILY REPORT

Connect daily reporting directly to data.

Do not ask staff to retype everything.

Auto-populate:

- date
- weather where integration available
- manpower
- equipment
- material deliveries
- work orders
- executed quantities
- inspections
- photographs

Engineer adds narrative and exceptions.

39. RFI

Current RFI capability exists but should be enhanced.

Workflow:

Draft
Submitted
Coordinator Review
Consultant Review
Additional Reviewer
Official Response
Contractor Acknowledgement
Closed

Fields:

- project
- location
- work package
- discipline
- subject
- question
- drawing references
- specifications
- photos
- due date
- responsible party
- schedule impact
- cost impact

Autodesk's current RFI implementation includes coordinator/reviewer roles, references to files, photos, submittals, schedules and other records.

40. SUBMITTALS

Types:

- material
- shop drawing

- sample
- method statement
- technical data
- mock-up
- test certificate

Workflow:

Subcontractor



Technical Office



PM



Consultant



Specialist Reviewers



Response

Responses:

A – Approved

B – Approved with Comments

C – Revise & Resubmit

D – Rejected

Keep revision history.

Autodesk currently supports submittal packages categorized by physical elements such as floors or rooms, which fits the location-based model proposed here.

41. DOCUMENT CONTROL

Create controlled registers:

- drawings
- specifications
- contracts
- reports

- technical documents
- correspondence
- method statements
- as-built documents
- O&M

Document fields:

Document Number

Title

Discipline

Type

Revision

Status

Project

Location

Package

Originator

Recipient

Issue Date

Review Date

42. REVISION CONTROL

Only one current revision.

Example:

STR-001

Rev 01 – Superseded

Rev 02 – Superseded

Rev 03 – Current

Superseded documents need a prominent visual warning.

The existing document library already maintains version history and resets approval when a new revision is uploaded.

Keep that logic and make it enterprise-grade.

43. TRANSMITTALS

Support:

- incoming
- outgoing
- recipients
- documents
- purpose
- acknowledgement
- response due date

Generate formal PDF transmittal automatically.

44. QA/QC

Modules:

ITP
WIR
MIR
Checklists
Tests
Hold Points
Witness Points
Quality Observations
NCR
Corrective Actions
Preventive Actions
Mock-ups
Calibration
Certificates
Punch Lists

45. WIR

Work Inspection Request.

Fields:

- project
- exact location
- BOQ item
- work package
- subcontractor
- activity
- inspection date
- checklist
- drawings
- method statement
- photographs

Workflow:

Site Engineer

→ QA/QC

→ PM where required

→ Consultant

Result:

Approved

Approved with Comments

Rejected

Reinspect

46. MIR

Material Inspection Request.

Link:

PO

Delivery

GRN

Material

Supplier

Submittal

Certificates

Test Results

Material should not become unrestricted usable inventory until the configured inspection status allows it.

47. NCR

Existing NCR functionality should remain but receive:

- location
- responsible party
- root cause
- corrective action
- preventive action
- cost impact
- schedule impact
- attachments
- verification

The current module already supports severity and state progression.

48. HSE

Add:

- induction
- toolbox talks
- JSA
- risk assessment
- permit to work
- hot work
- lifting
- excavation
- confined spaces
- observations
- near misses
- incidents

- PPE
- equipment inspections
- corrective actions
- emergency drills

Dashboard:

Man Hours

Days Without LTI

Open Observations

Overdue Safety Actions

Near Misses

Incidents

49. VARIATION / CHANGE MANAGEMENT

Workflow:

Change Event

↓

Potential Variation

↓

Technical Review

↓

Subcontractor/Supplier RFQ

↓

Cost Build-Up

↓

Schedule Impact

↓

Internal Approval

↓

Consultant Recommendation

↓

Client Approval

↓

Variation Order

Causes:

- design change
 - RFI response
 - client request
 - consultant instruction
 - unforeseen condition
 - authority
 - scope omission
-

50. VARIATION CALCULATIONS

Direct Cost

+ Labour

+ Equipment

+ Subcontract

+ Overhead

+ Profit

=

Variation Value

Track:

Submitted Value

Recommended Value

Approved Value

51. PAYMENT CERTIFICATES

Client side:

Previous Certified

Current Gross

Retention

Advance Recovery

Variations

Deductions

Tax

Net Certificate

Subcontractor side uses the same concept.

All certificates should produce formal generated documents.

52. RESOURCE / LABOUR

Track:

- employee
- subcontract labour
- discipline
- attendance
- project
- floor/location
- work package
- hours
- overtime
- cost rate

Calculation:

Labour Cost

=

Regular Hours × Rate

+

Overtime Hours × OT Rate

53. EQUIPMENT

Equipment master:

- code
- type
- serial
- ownership
- purchase/rental

- rate
- maintenance
- insurance
- certificates

Project usage:

Equipment

Hours

Location

Operator

Activity

Fuel

Cost

54. EQUIPMENT COST

Equipment Cost

=

Usage Hours × Internal/Rental Rate

plus configurable:

- fuel
- mobilization
- maintenance

55. MEETINGS

Meeting types:

- progress
- technical
- commercial
- safety
- coordination
- client
- consultant

Minutes contain:

Agenda
Attendees
Discussion
Decision
Action
Owner
Due Date

Actions automatically feed into the universal Action Center.

56. CORRESPONDENCE

Maintain formal:

- letters
- notices
- instructions
- claims
- contractual notifications

Every communication:

- numbered
 - revision-safe
 - searchable
 - linked to project records
 - auditable
-

57. UNIVERSAL ACTION SYSTEM

This is one of the most important shared components.

Create:

ActionItem

Fields:

Project
Source Type
Source ID
Location
Title
Assigned User
Assigned Organization
Priority
Due Date
Status
Description
Attachments
Comments
Reminder
Escalation

An action may originate from:

Site Visit
Observation
RFI
Submittal
Meeting
NCR
Inspection
Safety
Procurement
Material Shortage
Schedule Delay
Client Comment

Everyone gets one **My Actions** screen.

58. NOTIFICATION ENGINE

Channels:

- in-app

- email
- push
- optional SMS
- optional WhatsApp provider

Types:

- assignment
- due soon
- overdue
- approval requested
- approval completed
- rejected
- consultant comment
- material shortage
- delivery
- inspection
- payment
- expiring document

59. REMINDERS AND ESCALATION

Example:

Consultant Observation

Immediately

→ Notify PM

After 4h not acknowledged

→ Reminder

After 24h

→ Construction Manager

Due date missed

→ Projects Director

Everything configurable per project/template.

60. NOTIFICATION PREFERENCES

Users may configure:

Immediate
Daily Digest
Weekly Digest
Mute optional

Critical notices cannot be completely muted where company policy forbids it.

61. DASHBOARD — CEO / OWNER

Display company-level:

- total projects
 - contract backlog
 - revenue
 - collections
 - receivables
 - payables
 - cash requirement
 - total budgets
 - committed cost
 - actual cost
 - forecast cost
 - forecast profit
 - project margins
 - approved variations
 - pending variations
 - projects delayed
 - procurement exposure
 - overdue approvals
 - major quality issues
 - safety issues
-

62. DASHBOARD — PROJECTS DIRECTOR

- project health
 - PM performance
 - progress
 - schedule delay
 - commercial position
 - major RFIs
 - major NCRs
 - resource constraints
 - material shortages
 - major client decisions
-

63. DASHBOARD — CONSTRUCTION MANAGER

- today's site activities
 - lookahead
 - floors in progress
 - labour
 - equipment
 - inspections
 - consultant observations
 - blocked works
 - delays
 - safety
 - materials
-

64. DASHBOARD — PROJECT MANAGER

Main command centre:

My Actions

Project Progress
Milestones
Schedule
Quantities
Material Readiness
Procurement
Consultant Comments
RFIs
Submittals
WIR/MIR
NCR
Variations
Cost
Invoices
Subcontractors
Risks

65. DASHBOARD — SITE ENGINEER

Mobile-oriented:

My Work Today
Location
Current Drawings
Activities
Work Orders
Materials Available
Inspections
Consultant Comments
Photos
Daily Report
Notes

66. DASHBOARD — PLANNING ENGINEER

- baseline
 - actual
 - delayed activities
 - critical path
 - lookahead
 - resource curve
 - S-curve
 - SPI
 - activity updates
-

67. DASHBOARD — QS / COMMERCIAL

- BOQ
 - quantities
 - measurement
 - client valuation
 - subcontractor valuation
 - variations
 - commitments
 - budget
 - forecast
 - margin
-

68. DASHBOARD — PROCUREMENT

- PR pending
 - RFQ
 - quotations
 - comparisons
 - approvals
 - POs
 - delivery dates
 - shortages
 - delayed suppliers
-

69. DASHBOARD — STOREKEEPER

- receiving today
 - MIR pending
 - GRNs
 - stock
 - stock below minimum
 - reserved materials
 - issues
 - transfers
 - returns
 - waste
-

70. DASHBOARD — QA/QC

- inspections today
 - WIR pending
 - MIR pending
 - tests
 - NCR
 - overdue corrective actions
 - consultant quality comments
-

71. DASHBOARD — HSE

- permits
 - inspections
 - incidents
 - near misses
 - open observations
 - toolbox talks
 - overdue corrective actions
-

72. DASHBOARD — DOCUMENT CONTROLLER

- documents incoming
 - transmittals
 - review due
 - overdue reviews
 - new revisions
 - superseded drawings
 - correspondence
 - acknowledgement outstanding
-

73. DASHBOARD — FINANCE

- AR
 - AP
 - invoices
 - collections
 - due payments
 - project cashflow
 - retention
 - taxes
 - overdue invoices
-

74. DASHBOARD — CLIENT

Simple:

Progress
Milestones
Major Schedule
Photos
Client Decisions
Variations
Payment Status
Reports

Documents
Handover

75. DASHBOARD — CONSULTANT

Simple:

Project Progress
Milestones
Today's Inspection
WIR
MIR
RFIs
Submittals
Observations
NCR Closures
New Works Ready for Inspection
Latest Drawings
Daily Progress
Sticky Notes

76. DASHBOARD — SUBCONTRACTOR

My Scope
Today's Work
Drawings
Materials
Inspections
Comments
RFIs
Submittals
Quantities
Payment
Variations
Snags

77. REPORTING ENGINE

Every list/dashboard should support:

- filters
- saved views
- Excel
- PDF
- scheduled report
- email distribution
- charts
- drilldown

Reports must respect permissions.

78. PROJECT REPORT

Generate automatically:

Executive Summary

Progress

Schedule

Milestones

Manpower

Equipment

Quantities

Procurement

Materials

Quality

Safety

RFIs

Submittals

Consultant Observations

Variations

Financial Status

Photos

Key Risks

Actions

Weekly/monthly templates.

79. HANDOVER

Workflow:

Pre-Handover



Punch List



Rectification



Inspection



Testing & Commissioning



As-Built



O&M



Training



Taking Over

80. WARRANTY / DLP

Track:

- asset
 - location
 - supplier/subcontractor
 - warranty start
 - warranty end
 - defect
 - responsible party
 - SLA
 - rectification
 - approval
-

81. PROPERTY SALES

The current system includes buildings, units, reservations and contracts.

Keep this as an optional **Property Development module** rather than forcing it onto every construction company.

Expand eventually with:

- installment schedules
 - collections
 - broker commission
 - reservations
 - cancellations
 - handover
 - customer care
-

82. PRECONSTRUCTION / CRM

Company flow:

Lead

- Opportunity
 - Tender
 - Go/No-Go
 - Estimation
 - Supplier/Subcontract RFQs
 - Estimate
 - Proposal
 - Negotiation
 - Award
 - Contract
 - Project
-

83. TENDER / ESTIMATION

Support:

- cost library
- assemblies
- quantities
- subcontract quotations
- supplier quotations
- preliminaries
- overhead
- margin
- contingencies
- revisions

Calculation:

Direct Cost
 + Preliminaries
 + Overhead
 + Contingency
 + Profit
 =
 Tender Price

84. PROJECT TEMPLATES

Admin creates templates:

Residential Tower
 Villa Compound
 Commercial Building
 Infrastructure
 Fit-Out
 MEP
 Industrial

Template includes:

- WBS
- folders
- workflows
- roles
- approval rules

- document numbering
- QA checklists
- HSE checklists
- dashboard configuration
- notifications

Creating projects becomes much faster.

85. DOCUMENT TEMPLATES REQUIRED

The development agent should create configurable templates for:

Project

- Project Charter
- Project Team Sheet
- Project Execution Plan

Procurement

- PR
- RFQ
- Bid Comparison
- PO
- GRN
- MIR

Commercial

- Contract
- Subcontract
- Variation
- Payment Certificate
- Invoice

Technical

- RFI
- Submittal
- Transmittal
- Drawing Register

- Document Register
- Method Statement

Site

- Daily Report
- Site Instruction
- Visit Report
- Observation

Quality

- ITP
- WIR
- NCR
- Test Report
- Punch List

HSE

- Inspection
- JSA
- Permit
- Incident
- Toolbox Talk

Meetings

- MOM
- Action Register

Handover

- Handover Checklist
- Asset Register
- Warranty Certificate Register

Each should support company branding and PDF export.

86. AUTOMATIC DOCUMENT NUMBERING

Admin configures patterns.

Example:

PROJECT-DISCIPLINE-TYPE-SEQUENCE-REV

Output:

OSI-STR-RFI-0042
OSI-ARC-SUB-0018
OSI-QA-WIR-0231

87. SEARCH

Global search:

RFI-42
Tower A Floor 5
C35
ABC Contractor
Invoice 0082
Observation waterproofing

Search across permitted:

- records
 - documents
 - comments
 - projects
 - organizations
-

88. MOBILE / RESPONSIVE

Critical site functions must work well on mobile.

Prioritize:

- photos
 - daily report
 - inspection
 - WIR
 - MIR
 - consultant observations
 - action centre
 - drawings
 - material issues
 - labour
 - safety
-

89. OFFLINE MODE

Eventually support offline field work:

- cache latest drawings
- create observations
- fill inspections
- take photos
- complete checklist

Sync when online.

Conflict resolution must be explicit.

90. AUDIT LOG

Every material action must be immutable/auditable.

Record:

User

Organization

Timestamp

IP/session where appropriate

Action

Before

After
Project
Source

Especially:

- approvals
 - payments
 - contracts
 - quantities
 - document revisions
 - agent actions
-

91. WORKFLOW ENGINE

Do not hard-code separate logic for every process.

Create:

WorkflowTemplate
WorkflowStep
WorkflowInstance
WorkflowAction
Approval
EscalationRule

Steps support:

- user
 - role
 - organization type
 - amount condition
 - project condition
 - SLA
 - required attachments
 - required comments
 - parallel approval
 - sequential approval
-

92. API-FIRST ARCHITECTURE

Everything important should be accessible through an authenticated API.

Suggested API families:

/api/v1/companies

/api/v1/users

/api/v1/organizations

/api/v1/projects

/api/v1/projects/{id}/team

/api/v1/projects/{id}/locations

/api/v1/projects/{id}/wbs

/api/v1/boq

/api/v1/materials

/api/v1/inventory

/api/v1/procurement

/api/v1/contracts

/api/v1/subcontracts

/api/v1/versions

/api/v1/invoices

/api/v1/payments

/api/v1/rfis

/api/v1/submittals

/api/v1/inspections

/api/v1/observations

/api/v1/ncrs

/api/v1/documents

/api/v1/actions

/api/v1/notifications

/api/v1/reports

Procore's developer platform currently exposes construction objects such as RFIs, submittals, drawings and cost codes through OAuth-protected APIs; this is a good benchmark for making your own domain objects first-class API resources.

93. API REQUIREMENTS

Implement:

- OAuth/OIDC
- scoped API tokens
- service accounts
- pagination
- filters
- sorting
- versioning
- idempotency keys
- webhooks
- rate limiting
- API audit logs
- OpenAPI specification
- sandbox/test environment

Never expose direct unrestricted database access.

94. WEBHOOKS

Events:

project.created

rfi.created

rfi.overdue

submittal.submitted

observation.created

observation.closed

inventory.low

purchase_order.approved

delivery.received

invoice.created

invoice.overdue

payment.received

inspection.requested

inspection.completed

variation.approved

action.overdue

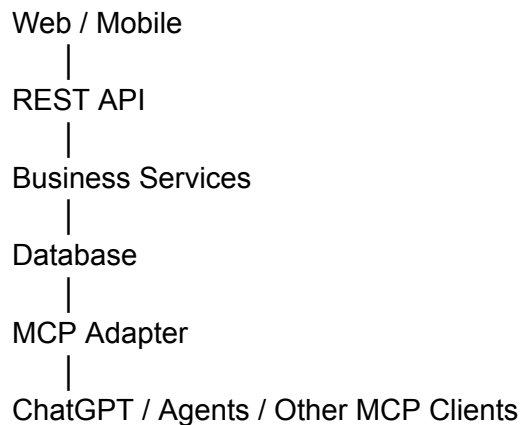
External automation tools can subscribe.

95. MCP SERVER

Build a **remote MCP server** over the same application service layer as the REST API.

Do not write business logic separately inside MCP.

Architecture:



As of September 2026, the MCP 2026-07-28 specification introduced a stateless protocol core, strengthened authorization and an extension model; legacy HTTP+SSE is on a deprecation path. Build against the current specification rather than older MCP tutorials.

96. MCP AUTHENTICATION

Use strong user authorization.

Agent permissions must inherit:

User
Organization
Project
Role

If a consultant asks:

What is Tower A margin?

the MCP layer should return:

Forbidden

because consultant permissions do not allow internal costs.

97. MCP READ TOOLS

Examples:

list_projects
get_project

get_project_progress
get_project_team
get_project_milestones

list_locations
get_floor_status

get_boq
get_quantities
calculate_remaining_quantities

get_material_requirements
get_inventory_status
get_material_shortages

list_rfis
get_rfi

list_submittals
list_inspections
list_observations
list_ncrs

get_project_cost_summary

list_invoices
list_payments

search_documents

get_my_actions

98. MCP WRITE TOOLS

Examples:

create_rfi
reply_to_rfi

create_site_observation
add_observation_comment
upload_observation_photo

create_purchase_requisition

create_inspection_request

update_daily_report

assign_action
complete_action

create_invoice_draft

create_variation_draft

99. HIGH-RISK AGENT ACTIONS

Do not allow autonomous financial/destructive execution without policy.

Examples requiring approval:

issue_purchase_order
approve_variation
send_client_invoice
record_payment
approve_subcontract_certificate
delete_document
close_project

The agent can prepare them automatically, but the configured authority must approve.

OpenAI's current ChatGPT documentation supports custom MCP applications with write/modify actions, subject to workspace permissions and confirmation behaviour, making this architecture suitable for direct ChatGPT integration where enabled.

The Apps SDK is also built on MCP, so the same integration can eventually provide interactive ChatGPT-native UI rather than only text tool calls.

100. AGENT EXAMPLES

Project Manager

User:

What do I need to handle today?

Agent retrieves:

My actions
Overdue RFIs
Consultant comments
Material shortages
Inspections
Schedule delay
Invoices

and returns prioritized work.

Procurement

User:

What materials will become short within the next 14 days?

Agent calculates:

schedule demand
inventory
reserved inventory
incoming PO
lead time

and returns shortages.

Then:

Create PRs for all critical shortages.

Agent creates draft PRs.

Consultant

Consultant:

Show everything awaiting my inspection on Tower A.

Agent returns permitted WIR/MIR records.

Consultant:

Approve WIR-331 with comment that curing must continue for 5 days.

Tool executes the approved action.

Owner

Which projects have the worst forecast margin?

Agent calculates portfolio results using permitted finance data.

101. AI ASSISTANCE INSIDE ERP

AI should assist—not silently control contractual records.

Potential features:

- summarize project status
- summarize consultant comments
- draft RFI
- suggest RFI categorization
- extract information from uploaded document
- summarize meeting minutes
- draft progress report
- detect overdue work
- identify material shortage
- identify unusual consumption
- compare BOQ vs actual
- flag commercial exposure
- summarize NCR trends
- search documents conversationally

Autodesk currently allows AI-assisted creation of RFI fields but still requires the user to review generated information before relying on it; that human-review principle should also apply here.

102. DATA MODEL — CORE ENTITIES

At minimum:

Company
Department
User
Employee

Role
Permission

Organization
OrganizationUser

Project
ProjectParticipant
ProjectTeamMember

ProjectLocation
LocationType

WBS
ScheduleActivity
Milestone
WorkPackage

BOQSection
BOQItem
BOQLocationAllocation

Material
MaterialRecipe
MaterialRequirement

Warehouse
InventoryStock
StockReservation
StockMovement

PurchaseRequisition
RFQ
RFQVendor
SupplierQuotation
BidComparison
PurchaseOrder
Delivery
GRN

Contract
ContractLine
Subcontract
SubcontractLine

Variation
PaymentCertificate

Invoice
Payment
Expense

DailyReport
SiteVisit
Observation
ObservationComment

RFI
RFIResponse
Submittal
SubmittalRevision

ITP
WIR
MIR
Inspection
Test
NCR

Document
DocumentRevision
Transmittal
Correspondence

Meeting
MeetingAction

Equipment
EquipmentUsage
Timesheet

ActionItem
Notification

WorkflowTemplate
WorkflowInstance
WorkflowStep

AuditEvent

103. EMPTY-STATE STANDARD

This is mandatory.

Every module must work with zero records.

Required states:

Loading
Empty
Populated
Error
Permission Denied
Offline

No module may assume data exists.

This directly solves the requirement that:

Client dashboard should work even before a client exists.

Same for consultant dashboard and all other views.

104. USER EXPERIENCE STANDARD

UI should follow:

Executive

minimal + KPI-driven.

Office team

dense + efficient.

Site/mobile

large actions + minimal typing.

Consultant

review-oriented.

Client

simple + executive.

Subcontractor

scope/action-oriented.

Avoid exposing 30 menu items to every person.

105. GLOBAL COMMAND CENTRE

Top bar should provide:

- Search
- My Actions
- Notifications
- Quick Create
- Project Switcher
- Help
- Profile

Quick Create changes according to user role.

106. PROJECT SWITCHER

When a user belongs to multiple projects:

- Project A
- Project B
- Project C

Selecting project scopes the whole workspace.

No accidental cross-project data.

107. DEVELOPER/AGENT BUILD SEQUENCE

The following sequence should be followed.

Do **not** jump randomly between modules.

PHASE 1 — Audit the Existing Codebase

Agent must:

1. Inspect frontend.
2. Inspect backend.
3. Inspect database.
4. Inspect routes.
5. Inspect API.
6. Inspect authentication.
7. Inspect current permissions.
8. Map every current table.
9. Map every current module.
10. Map every current workflow.

Deliver documents:

- `CURRENT_SYSTEM_ARCHITECTURE.md`
- `CURRENT_DATABASE_SCHEMA.md`
- `CURRENT_MODULE_MATRIX.md`
- `CURRENT_API_MATRIX.md`
- `CURRENT_PERMISSION_MATRIX.md`
- `GAP_ANALYSIS.md`

Acceptance:

No new architecture work starts until existing functionality is mapped.

PHASE 2 — Core Data Architecture

Build:

- organizations
- participants
- enhanced roles
- project locations
- WBS
- work packages
- cost codes

Documents:

- [TARGET_DATA_MODEL.md](#)
- [ENTITY_RELATIONSHIP_DIAGRAM.md](#)
- [MIGRATION_PLAN.md](#)

Acceptance:

Existing records migrate without loss.

PHASE 3 — Permission Engine

Build:

- role permissions
- project permissions
- organization access
- financial restrictions

Documents:

- [RBAC_MATRIX.md](#)
- [PERMISSION_TEST_CASES.md](#)

Acceptance:

Automated security tests prove external parties cannot access internal data.

PHASE 4 — Project Creation Wizard

Build the full 11-step project wizard.

Documents:

- [PROJECT_CREATION_SPEC.md](#)
- [PROJECT_TEMPLATE_SPEC.md](#)

Acceptance:

Creating one project produces:

- locations
- workspace
- permissions
- team
- folders
- workflows
- dashboards

without manual database configuration.

PHASE 5 — Universal Workflow Engine

Build configurable workflows.

Documents:

- [WORKFLOW_ENGINE.md](#)
- [APPROVAL_MATRIX.md](#)

Acceptance:

RFI, Submittal and purchase approval can all use the same engine.

PHASE 6 — Action / Reminder / Notification Engine

Build:

- actions
- reminders
- escalation
- notifications
- digests

Documents:

- `NOTIFICATION_RULES.md`
- `ESCALATION_MATRIX.md`

Acceptance:

All important records can create actionable tasks.

PHASE 7 — Location + Quantity Engine

Build:

- site/building/floor/area
- BOQ allocations
- physical progress
- roll-ups

Documents:

- `QUANTITY_CALCULATION_RULES.md`
- `LOCATION_MODEL.md`

Acceptance:

User can calculate quantities from floor → building → project.

PHASE 8 — Material Planning

Build recipes and demand.

Documents:

- `MATERIAL_RECIPE_SPEC.md`
- `MATERIAL_DEMAND_FORMULAS.md`

Acceptance:

Selecting a floor/activity correctly calculates its material demand.

PHASE 9 — Inventory

Build:

- warehouse
- stock
- reserve
- issue
- transfer
- return
- waste

Documents:

- `INVENTORY_RULES.md`

Acceptance:

Every unit of inventory can be traced.

PHASE 10 — Auto-Replenishment

Build:

- min/max
- safety
- lead time
- reorder point
- projected stock
- auto PR

Documents:

- [REPLENISHMENT_FORMULAS.md](#)
- [AUTOPURCHASE_POLICY.md](#)

Acceptance:

Test shortage automatically creates correct draft procurement action.

PHASE 11 — Procurement

Implement complete PR → PO → GRN flow.

Documents:

- [PROCUREMENT_WORKFLOW.md](#)
- document templates.

Acceptance:

Full procurement test completes without manual database editing.

PHASE 12 — Commercial / Contracts

Implement:

- client contract
- subcontract
- variations
- commitments
- payment certificates

Documents:

- [COMMERCIAL_RULES.md](#)
- [CONTRACT_CALCULATIONS.md](#)

Acceptance:

Budget, commitments, actual and forecast reconcile correctly.

PHASE 13 — Finance & Invoicing

Implement:

- AR
- AP
- client invoices
- supplier invoices
- collections
- reminders

Documents:

- [FINANCIAL_CALCULATION_RULES.md](#)
- [INVOICE_WORKFLOW.md](#)

Acceptance:

Invoice/payment scenarios reconcile exactly.

PHASE 14 — Consultant Portal

Implement:

- login
- dashboard
- inspections
- observations
- RFIs
- submittals
- drawings
- notes

Documents:

- [CONSULTANT_PORTAL_SPEC.md](#)

Acceptance:

Consultant can complete their daily role without entering internal ERP screens.

PHASE 15 — Client Portal

Implement simple owner dashboard.

Documents:

- [CLIENT_PORTAL_SPEC.md](#)

Acceptance:

Works with:

- client
 - no client
 - no records
 - full project.
-

PHASE 16 — Subcontractor Portal

Implement trade-partner workflow.

Documents:

- [SUBCONTRACTOR_PORTAL_SPEC.md](#)
-

PHASE 17 — QA/QC

Implement ITP/WIR/MIR/NCR.

Documents:

- [QAQC_WORKFLOW.md](#)
 - templates/checklists.
-

PHASE 18 — HSE

Implement full safety system.

Documents:

- [HSE_WORKFLOW.md](#).
-

PHASE 19 — Document Control

Implement:

- register
- revisions
- transmittals
- superseded protection
- correspondence

Documents:

- [DOCUMENT_CONTROL_STANDARD.md](#)
 - [NUMBERING_STANDARD.md](#).
-

PHASE 20 — Planning

Implement project schedule + lookaheads.

Documents:

- [PLANNING_CALCULATIONS.md](#).
-

PHASE 21 — Dashboards

Create every role dashboard listed earlier.

Documents:

- [DASHBOARD_MATRIX.md](#).

Acceptance:

Login as every role and verify the dashboard.

PHASE 22 — Reports

Create:

- daily
- weekly
- monthly
- commercial
- procurement
- management

Documents:

- `REPORT_CATALOG.md`.
-

PHASE 23 — API

Build versioned external API.

Documents:

- OpenAPI specification
- integration guide
- authentication guide

Acceptance:

Automated API tests.

PHASE 24 — MCP

Build remote MCP server.

Documents:

- `MCP_SERVER_SPEC.md`
- `MCP_TOOL_CATALOG.md`
- `MCP_PERMISSION_MAP.md`

- MCP_SECURITY.md

Acceptance:

External MCP client can securely:

Read Project
Read Actions
Create RFI
Create Draft PR
Add Observation Comment

according to permissions.

PHASE 25 — Agent Layer

Build agent workflows:

PM Assistant
Procurement Assistant
Commercial Assistant
Consultant Assistant
Executive Assistant
Document Assistant

Every agent uses the same API/MCP security model.

PHASE 26 — Mobile Optimization

Test all field functions.

PHASE 27 — Automated Testing

Must include:

Unit tests

calculations.

Integration tests

workflow connections.

Permission tests

every role.

E2E

complete construction scenarios.

Financial reconciliation

all amounts.

Empty-state tests

zero-data projects.

Mobile tests

site flows.

Agent tests

MCP permissions/actions.

108. MASTER END-TO-END TEST PROJECT

Developer should seed one realistic project.

Example:

Residential Towers
2 Buildings
12 Floors Each

4 Units/Floor
Multiple Subcontractors
Consultant
Client
Project Team
Warehouse
BOQ
Schedule

Run complete scenario:

Tender
↓
Award
↓
Project Creation
↓
Team
↓
BOQ
↓
Budget
↓
Quantities
↓
Material Requirement
↓
PR
↓
PO
↓
Delivery
↓
MIR
↓
GRN
↓
Stock
↓
Issue
↓
Work
↓
WIR

↓
Consultant Observation
↓
Rectification
↓
Consultant Approval
↓
Progress
↓
Client Certificate
↓
Invoice
↓
Collection
↓
Variation
↓
Subcontract Payment
↓
Handover
↓
Warranty

The current guide already provides an end-to-end test scenario through projects, BOQ, work orders, site, quality, documents, invoices and unit sales; retain that idea and extend it into the company-wide scenario above.

109. DEFINITION OF DONE FOR EVERY MODULE

A module is **not finished because the page loads**.

Every module must have:

- ✓ Database
- ✓ API
- ✓ UI
- ✓ Mobile/responsive UI
- ✓ Empty state

- ✓ Loading state
 - ✓ Error state
 - ✓ Permissions
 - ✓ Audit log
 - ✓ Search
 - ✓ Filters
 - ✓ Export
 - ✓ Attachments
 - ✓ Comments if applicable
 - ✓ Notifications
 - ✓ Reminders
 - ✓ Workflow
 - ✓ Dashboard widgets
 - ✓ Reports
 - ✓ MCP read tools
 - ✓ MCP write tools where safe
 - ✓ Automated tests
 - ✓ Documentation
-

110. ZERO-BROKEN-VIEW POLICY

It is impossible to guarantee that software will literally never contain a bug.

What can and should be required is:

No uncaught UI exceptions

No blank pages

No unresolved promises

No null/undefined rendering

No unhandled API errors

No missing permission handling

No route without fallback

No hidden financial exposure

No record without project scope where project scope is required

Use:

- error boundaries
- structured API errors
- monitoring

- logs
 - health checks
 - automated tests
 - database constraints
 - transaction handling
-

111. FINAL PRODUCT PRINCIPLE

The construction ERP should not feel like hundreds of forms.

For every person, it should feel like:

What do I need to do today?

What is waiting on me?

What is late?

What changed?

What needs approval?

What is blocked?

What happens next?

while management can answer:

Where are we?

Are we ahead or behind?

What have we spent?

What have we committed?

What will it cost to finish?

What do clients owe us?

What do we owe others?

What materials will run out?

What approvals are blocking work?

What risks need attention?

And an authorized AI agent should be able to answer the same questions and perform permitted actions securely.

112. TARGET TECHNOLOGY POSITION

The desired architecture should borrow the strongest concepts rather than copy one application:

- **Procore-style:** connected project objects, field execution, integrations and construction-specific APIs.
- **Autodesk-style:** configurable RFI/submittal workflows, linked issues/records and project collaboration.
- **Oracle Aconex-style:** rigorous documents, organizations, workflows and auditability.
- **Primavera Unifier-style:** enterprise cost/project controls.
- **Modern MCP/API layer:** first-class agent access rather than screen scraping. The current MCP specification is designed around stateless, routable servers with hardened authorization, and OpenAI currently supports custom MCP-powered apps and write actions in supported workspace configurations.

The result should therefore be:

One construction company operating system. One source of truth. Every project has its own team, permissions, files, quantities, materials, contracts, invoices, workflows and dashboards. Every internal and external party receives only the interface and information relevant to them. Every major record is location-linked, auditable and actionable. The entire backend is API-first and MCP-ready so ChatGPT or another authorized agent can operate the ERP safely.